CARDSTOCK ROCKET FLIGHT FORENSICS & TEARDOWN LAB

PART 1: PRECISION DISSECTION, TOLERANCES & TECHNICAL BLUEPRINT DRAFTING

Orange High School • Unit 2 Physics
Standards: HS-PS2-1 HS-ETS1-2

STUDENT NAME: _____ **PERIOD:** _____ **DATE:** _____ **ROCKET CALL SIGN:** _____

🔬 **AEROSPACE ACCIDENT INVESTIGATION & REVERSE-ENGINEERING:** Carefully dissect your flown rocket to document as-built dimensions, fin shape, and stability points.

⚖️ TABLE 1: FLOWN ROCKET TEARDOWN & MASS BUDGET AUDIT

RECORD VIA METRIC RULER & DIGITAL BALANCE (0.1 G PRECISION)

Component	Measured Quantity	Recorded Value	Physical Significance
1. Fuselage Cylinder	Length (L) & Outer Diameter (D)	L = _____ cm | D = _____ cm (½" pipe ≈ 1.3 cm)	Pneumatic stroke volume & body drag
2. Fuselage Mass & Integrity	Dry Fuselage Mass & Condition	Mass: _____ g | [ ] Pristine [ ] Crimped	Pressure containment vs dead weight
3. Conical Nose Cone	Nose Height & Tip Seal	H = _____ cm | Added Tip Mass (if any): _____ g (Material: Standard Cardstock & Tape)	Aerodynamic profile & airtight seal
4. Fin Geometry	Fin Count, Height & Width	____ fins | Height: ____ cm | Width: ____ cm Alignment: [ ] Flush with tail [ ] Shifted _____ cm up from tail	Aerodynamic stability & flight path
5. Total As-Flown Mass	Fully Assembled Mass (g & kg)	m = _____ g = _____ kg	Newton's 2nd Law acceleration (a = F/m)
6. Center of Mass (CM)	Balance Point from Nose Tip	CM = _____ cm from nose tip	Physical balance pivot point

📐 SECTION B: TECHNICAL DRAFTING BLUEPRINT & FIN OUTLINE (1 CM × 1 CM GRID)

ENGINEERING DRAFTING

Drafting Requirements: (1) Draw your exact hand-cut fin shape and rocket elevation along the centerline using the 1 cm × 1 cm grid, (2) Include dimension extension lines with arrows for Fuselage Length, Nose Height, and Fin Width & Height, (3) Mark and label the Center of Mass (⊕ CM) and Center of Pressure (⊙ CP).

TOP VIEW / SIDE ELEVATION GRID (GRID SCALE: 1 SQUARE = 1 CM × 1 CM)

NOSE TIP (LEFT) → → → TAIL & FINS (RIGHT)

Stability Check: Is CM forward of CP? [ ] YES (Stable) [ ] NO (Tumble Hazard) **Stability Margin (CP - CM):** _____ cm (Target: ≥ 1.3 cm / 1 body diameter [½" pipe])

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Student: _____ **Period:** _____

PART 2: SATELLITE RECOVERY TELEMETRY, TRAJECTORY PHYSICS & CER DEFENSE

Launcher GPS: Period 0: 33°47'50.6"N 117°50'48.7"W | **Periods 4–6:** 33°47'48.8"N 117°50'48.4"W (Campus Quad) Launcher Pressure: **Calibrated 80–100 PSI**

🛰️ SECTION C: SATELLITE RECOVERY TELEMETRY (DIGITAL CAMPUS MAP PIN)

UNIT_2/ROCKET_RECOVERY/INDEX.HTML

1. Ground Distance (d):

_____ **meters**

(Multiply meters × 3.28 = _____ ft)

2. Flight Azimuth (Bearing):

_____**° (Compass Dir: _____)**

(0° = North, 90° = East, 180° = South)

3. Touchdown Location / Landmark:

Specific landing spot (e.g. Quad grass, roof, breezeway):

4. Post-Landing Structural Health:

[ ] Pristine / Re-flyable

[ ] Bent Fins / Accordion Nose

🔍 SECTION D: AERODYNAMIC FLIGHT FORENSICS ANALYSIS

PHYSICS DIAGNOSTIC

1. Fuselage Fit & Pressure Seal: Airtight Seal vs. Launch Energy

How snug was your rocket's fit on the 0.5" launch pipe? Explain whether air pressure leaked past a loose fuselage (reducing launch velocity), if excessive friction slowed it down, or if a snug seal converted maximum 80–100 PSI pressure into kinetic energy:

2. Flight Behavior & Drift Forensics: Why Did It Travel Across Campus?

Circle behavior: [ ] Vertical climb & wind drift [ ] Long arching flight [ ] Corkscrew spin [ ] Hard bank [ ] Tumble [ ] Blowout. How did your fin shape/position or launcher angle affect flight, spin, or stability?

📝 SECTION E: CLAIM-EVIDENCE-REASONING (CER) FLIGHT FORENSICS SYNTHESIS

SUMMATIVE EVALUATION

Prompt: Formulate an engineering claim evaluating your rocket's flight performance and touchdown location. Defend which design variable (fuselage pressure seal, fin shape & alignment, or launch angle/wind) was most decisive for your flight.

- **Claim:** Direct statement explaining the primary factor governing your rocket's flight path and landing location.
- **Evidence:** Cite quantitative teardown data (Ground Distance in m/ft, Flight Azimuth, Fin Dimensions in cm, Fuselage Fit, Stability Margin in cm).
- **Reasoning:** Explain the physics connecting your evidence to your claim using Newton's 2nd Law ($a = F_{net} / m$) or aerodynamic torque stability (CM vs CP).